

SECTION 16055

OVER-CURRENT PROTECTIVE DEVICE COORDINATION

PART 1 GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes computer-based, fault-current and over-current protective device coordination studies, and the setting of these devices.

1.3 SUBMITTALS

- A. Qualification Data: For coordination-study specialist.
- B. Other Action Submittals:
 - 1. Coordination-study input data, including completed computer program input data sheets.
 - 2. Coordination-study report.
 - 3. Equipment evaluation report.
 - 4. Setting report.
- C. QUALITY ASSURANCE
 - 1. Software algorithms shall comply with requirements of standards and guides specified in this Section. Manual calculations are not acceptable.
 - 2. Coordination-Study Specialist Qualifications: An organization experienced in the application of computer software used for studies, having performed successful studies of similar magnitude on electrical distribution systems using similar devices.
- D. Comply with IEEE 399 (Brown Book) for general study procedures.

- E. Comply with IEEE 242 (Buff Book) for short-circuit currents and coordination time intervals.

PART 2 PRODUCTS

2.1 COMPUTER SOFTWARE

- A. Computer Software: Subject to compliance with requirements, provide computer software program by SKM Systems Analysis, Inc.

2.2 COMPUTER SOFTWARE PROGRAM REQUIREMENTS

- A. Comply with IEEE 399 (Brown Book).
- B. Computer software program shall be capable of plotting and diagramming time-current-characteristic curves as part of its output. Computer software program shall report device settings and ratings of all over-current protective devices.

PART 3 EXECUTION

3.1 EXAMINATION

- A. Examine Project over-current protective device submittals for compliance with electrical distribution system coordination requirements and other conditions affecting performance. Protective devices to be coordinated are as follows:
 - 1). All circuit breakers and fuses from the main switchboard (or main transformer's secondary protection) to feeder breakers in the distribution panels.
 - 2). Ground fault protection in the main switchboard shall be coordinated to clear a 40A breaker in 480/277V branch panels.
- B. Proceed with coordination study only after relevant equipment submittals have been approved.

3.2 FAULT-CURRENT STUDY

- A. Source Impedance: Utility fault-current contribution as furnished by owner.
- B. Study electrical distribution system from normal and alternate power sources throughout electrical distribution system for Project and use approved computer software program to calculate values. Include studies of system-switching configurations and alternate operations that could result in maximum fault conditions.

- C. Calculate momentary and interrupting duties on the basis of maximum available fault current.
- D. Calculations to verify interrupting ratings of over-current protective devices shall comply with the following:
 - 1). Low-Voltage Circuit Breakers: IEEE 1015 and IEEE C37.50.
 - 2). Low-Voltage Fuses: IEEE C37.46.
 - 3). Circuit Breakers: IEEE C37.13.
- E. Study Report: Enter calculated X/R ratios and interrupting (5-cycle) fault currents on electrical distribution system diagram of the report. List other output values from computer analysis, including momentary (1/2-cycle), interrupting (5-cycle), and 30-cycle fault-current values for 3-phase, 2-phase, and phase-to-ground faults.
- F. Equipment Evaluation Report: Prepare a report on the adequacy of over-current protective devices and conductors by comparing fault-current ratings of these devices with calculated fault-current momentary and interrupting duties.

3.3 COORDINATION STUDY

- A. Gather and tabulate the following input data to support coordination study:
 - 1). Product Data for over-current protective devices specified in other Division 16 Sections and involved in over-current protective device coordination studies. Use equipment designation tags that are consistent with electrical distribution system diagrams, over-current protective device submittals, input and output data, and recommended device settings.
 - 2). Electrical distribution system diagram showing the following:
 - a). Circuit-breaker and fuse-current ratings and types.
 - b). Relays and associated power and current transformer ratings and ratios.
 - c). Transformer kilovolt amperes, primary and secondary voltages, connection type, impedance, and X/R ratios.
 - d). Generator kilovolt amperes, size, voltage, and source impedance.
 - e). Cables. Indicate conduit material, sizes of conductors, conductor insulation, and length.
 - f). Busway, ampacity and impedance.
 - g). Motor horsepower and code letter designation according to NEMA MG 1.

- 3). Data sheets to supplement electrical distribution system diagram, cross-referenced with tag numbers on diagram:
 - a). Special load considerations, including starting inrush currents and frequent starting and stopping.
 - b). Magnetic inrush current overload capabilities of transformers.
 - c). Motor full-load current, locked rotor current, service factor, starting time, type of start, and thermal-damage curve.
 - d). Ratings, types, and settings of utility company's over-current protective devices.
 - e). Special over-current protective device settings or types stipulated by utility company.
 - f). Time-current-characteristic curves of devices indicated to be coordinated.
 - g). Manufacturer, frame size, interrupting rating in amperes rms symmetrical, ampere or current sensor rating, long-time adjustment range, short-time adjustment range, and instantaneous adjustment range for circuit breakers.
 - h). Manufacturer and type, ampere-tap adjustment range, time-delay adjustment range, instantaneous attachment adjustment range, and current transformer ratio for over-current relays.
 - i). Panelboards, switchboards, motor-control center ampacity, and interrupting rating in amperes rms symmetrical.
- B. Perform coordination study and prepare a written report using the results of fault-current study and approved computer software program. Comply with IEEE 399.
- C. Comply with NFPA 70 for over-current protection of circuit elements and devices.
- D. Comply with IEEE 242 (Buff Book) recommendations for fault currents and time intervals.
- E. Overcurrent protective devices shall be set in order to achieve arc flash PPE level 0 or 1 for branch panels and PPE 0, 1 or 2 for distribution panels.
- F. Transformer Primary Over-current Protective Devices:
 - 1). Device shall not operate in response to the following:
 - a). Self-cooled, full-load current or forced-air-cooled, full-load current, whichever is specified for that transformer.
 - b). Permissible transformer overloads according to IEEE C57.96 if required by unusual loading or emergency conditions.

- 2). Device shall protect transformer according to IEEE C57.12.00, for fault currents.
- G. Motors served by voltages over 600 V shall be protected according to IEEE 620.
- H. Conductor Protection: Protect cables against damage from fault currents according to ICEA P-32-382, ICEA P-45-482, and conductor melting curves in IEEE 242 (Buff Book). Verify adequacy of phase conductors at maximum three-phase bolted fault currents, equipment grounding conductors, and grounding electrode conductors at maximum ground-fault currents.
- I. Coordination-Study Report: Prepare a written report indicating the following results of coordination study:
- 1). Tabular Format of Settings Selected for Over-current Protective Devices:
 - a). Device tag
 - b). Relay-current transformer ratios; and tap, time-dial, and instantaneous-pickup values.
 - c). Circuit-breaker sensor rating; and long-time, short-time, and instantaneous settings.
 - d). Fuse-current rating and type.
 - e). Ground-fault relay-pickup and time-delay settings.
 - 2). Coordination Curves: Prepared to determine settings of over-current protective devices to achieve selective coordination. Graphically illustrate that adequate time separation exists between series devices, including power utility company's upstream devices. Show the following specific information:
 - a). Device tag
 - b). Voltage and current ratio for curves.
 - c). Three-phase and single-phase damage curves for each transformer.
 - d). No damage, melting, and clearing curves for fuses.
 - e). Circuit Breaker trip characteristic curves.
 - f). 12kV cable damage curves.
 - g). Transformer inrush points.
 - h). Maximum fault-current cutoff point.
 - 3). Evaluation: Note areas of lack of coordination and how they may or may not affect selectivity. If changes to the electrical system can improve coordination, make recommendations of what changes should be made.

3.4 OVERCURRENT PROTECTIVE DEVICE SETTING

A. Owner will perform device settings.

OR

B. Device settings will be done as specified in other Division 16 specifications.
Engage a qualified testing agency to perform the following device setting and to prepare test reports.

END OF SECTION